Oracle Bone Reassembly Based on Diffusion Model

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Oracle bones were primarily cattle scapulae and tortoise plastrons, as well as other animal bones, used for divination during the late Shang dynasty (ca. 1200–1045 B.C.) in ancient China. The inscriptions on them, which represent the earliest known form of the Chinese writing system, provide invaluable insights into early Chinese language and history. Since their rediscovery in 1899, over 150,000 oracle bones have been excavated (Jiang 2019: 191). However, many of them have become fragmented due to various factors, including the divination process, natural deterioration, and human handling. Therefore, reassembling digital images of these fragments significantly contributes to reconstructing the original inscriptions and advancing our understanding of ancient Chinese civilisation.

Traditionally, this reassembly task was performed manually by scholars, which was both labour-intensive and time-consuming. Given the visual nature of the task, it is particularly amenable to computer vision techniques. In recent years, computer-assisted methods have emerged for this task. However, the existing methods rely on either basic computer visualisation techniques or traditional image processing, with no integration of machine learning, resulting in limited sophistication and low efficiency.

To address this issue, we propose a novel approach based on Diffusion Models, a class of generative deep learning architectures that have demonstrated remarkable performance in computer vision tasks in recent years. Although their potential application to archaeology and cultural heritage studies has been acknowledged in the computer science literature, few studies have thoroughly explored this

domain. Specifically, this paper adopts the method of Denoising Diffusion Probabilistic Models (Ho et al. 2020).

The dataset used in this study consists of images cropped from published books featuring oracle bone rubbings. Firstly, we introduce a numerical interpretation of fragment data incorporating features, such as location, rotation, and edges. We regard the fragmented oracle bones as poses with noise, then the reassembly process aims to denoise. In the training process, we add Gaussian noise incrementally to the fragment data, and then train a neural network to reverse the process and obtain the initial pose. During the reverse diffusion process, the model reconstructs the original poses of the oracle bones. Furthermore, we add additional weight on the feature points at the edges of the oracle bones to optimise the model's performance.

We tested our model on oracle bones that had previously been reassembled by scholars, confirming our model's ability to rediscover the matches. Experimental results showed that our method significantly improved the efficiency over prior methods, such as feature-point matching method and some methods based on object detection. We also applied our model to a large, general dataset of oracle bones, where its new discoveries still require validation by human experts.

The limitation of our model is that we primarily focus on the visual shape of the oracle bone fragments and inscriptions, while disregarding the semantic meaning of the inscriptions and the divination marks, which may affect the reliability of the results. Therefore, it is necessary for human experts to examine the reassemblies to ensure their correctness.

Our research offers a promising computational approach to assist scholars in reassembling oracle bone fragments. By enabling accurate and large-scale reconstruction of oracle bone inscriptions, it

supports the decipherment of the inscriptions and the classification and periodisation of oracle bones. We anticipate that this approach will make a substantial contribution to the interdisciplinary study of artificial intelligence, archaeology and early Chinese history, and foster new methods in this emerging field.

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